



A comprehensive experimental analysis of gadolinium active magnetic regenerators



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HIGHLIGHTS

- ▶ A new experimental device for active magnetic regenerators (AMR) was developed.
- ▶ An experimental analysis of six different gadolinium AMRs was performed.
- ▶ Best characteristics were obtained by the parallel-plate AMR (0.1 mm of spacing).
- ▶ Almost 20 K of the temperature span was measured.

ARTICLE INFO

Article history:

Received 21 August 2012

Accepted 4 January 2013

Available online 24 January 2013

Keywords:

Magnetic refrigeration

Active magnetic regenerator

Gadolinium

Experiment

Magnetocaloric effect

Permanent magnet

ABSTRACT

The main goal of this study was an experimental comparison of six different active magnetic regenerators (AMRs) with gadolinium as the magnetocaloric material. The analysis was carried out for three different parallel-plate AMRs (with different porosities and different orientations of the plates in the magnetic field) and three different packed-bed AMRs (filled with spheres, powders and cylinders). Since the operation of an AMR is strongly affected by the operating conditions, the experiments were performed at different mass-flow rates and at different operating frequencies. These were required in order to define the optimum corresponding operating conditions for the analyzed AMRs. As comparative criteria the maximum temperature span, the cooling capacity and the experimentally predicted *COP* were taken into consideration.

The experimental analysis was performed on a new prototype of magnetic refrigerator designed as an experimental device. Its operation is based on the linear movement of a permanent-magnet assembly over a static AMR. The magnet assembly provides a measured magnetic field of about 1.15 T.

The results reveal that the geometry of the AMR (the form of the magnetocaloric material) has a crucial impact on the performance of the magnetic refrigerator. The best overall cooling characteristics (temperature span, cooling capacity and *COP*) were obtained for the parallel-plate AMR with the smallest porosity (~25%) and the orientation of the plates parallel to the magnetic field. This particular AMR generated a temperature span of 20 K, which is also the largest, so-far measured and published temperature span with a parallel-plate AMR for a given magnetic field change generated by permanent magnets. With respect to the comparison of the experimentally predicted *COP* values, the parallel-plate AMRs show higher efficiencies than the packed-bed AMRs.

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1. Introduction

Magnetic refrigeration at room temperature is a new, emerging technology that might, with further development, represent an alternative to the current, widely used, vapour-compression technology. Magnetic refrigeration is based on the magnetocaloric effect, which was discovered by Warburg in 1881 in iron [1]. In subsequent years a number of materials with more pronounced

magnetocaloric effects were discovered, but applications were limited to cryogenic temperatures. The development of magnetic refrigeration at room temperature began in 1976 when Brown developed the first prototype of a magnetic refrigerator working at room temperatures [2]. In this prototype gadolinium was used as the magnetocaloric material. Since then gadolinium has become the reference magnetocaloric material in the scientific community. Significant progress in the development of magnetocaloric materials with a Curie temperature near room temperature was made in 1997, when Gschneidner and Pecharchy [3] presented the discovery of the so-called giant magnetocaloric effect in the $Gd_5Si_2Ge_2$ alloy.

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Nomenclature

A_{ht}	total heat transfer area [m ²]
COP	coefficient of performance [–]
d	sphere diameter [m]
d_h	hydraulic diameter [m]
ε	porosity [–]
L	length [m]
m_s	mass of magnetocaloric material [kg]
$\dot{m}_f$	mass-flow rate [kg/s]
ν	operation frequency [Hz]
P_f	fluid-flow period duration (in one direction) [s]
$\dot{Q}_c$	cooling capacity [W]
ρ_f	fluid density [kg/m ³]
s	specific entropy [J/kg K]
t	plate thickness [m]
T	temperature [K]
ΔT	temperature span [K]
V^*	ratio of displaced fluid volume [–]
V_{AMR}	volume of the AMR [m ³]
$\dot{W}_{mag}$	power of the magnetic work [W]

In the past fifteen years, a number of different alloys with a giant magnetocaloric effect were discovered. Among them, the greatest potential is shown by alloys based on LaFeSi and MnFeP [4]. However, it is not clear which material will be established as the most appropriate. Namely, the major problem with these sintered materials is the difficulty in manufacturing them into fine, porous geometries for more efficient heat transfer.

It is important to point out that existing magnetocaloric materials do not have a sufficiently large magnetocaloric effect to be directly implemented in a refrigeration system. Therefore, in the magnetic-refrigeration thermodynamic cycle a heat-regeneration process is required in order to increase the temperature span of the device. Among the different types of heat regenerators, in recent years the active magnetic regenerator (AMR) has shown the greatest potential for an exploitation of the magnetocaloric effect.

An AMR is a porous structure of a magnetocaloric material through which a heat-transfer fluid is pumped in the counter-flow direction. The AMR has a double function in a magnetic refrigerator. It works as the refrigerant (it contains the magnetocaloric material) as well as the heat regenerator. All the prototypes of magnetic refrigerators developed in recent years have been based on an AMR [5]. The operation of an AMR is based on four steps, which are in general analogue to the vapour-compression refrigeration processes. These steps are (Fig. 1):

- Magnetization:** AMR is exposed to the change – the increase of the magnetic field – consequently each piece of the magnetocaloric material in the AMR is heated up.
- Fluid flow** from the cold external heat exchanger (Fig. 1 – CHEX on the left-hand side) through the heated magnetocaloric material into the hot external heat exchanger (Fig. 1 – HHEX on the right-hand side) where the heat received from the magnetocaloric material is transmitted to the surroundings.
- Demagnetization:** AMR is exposed to the change – the decrease of the magnetic field – and consequently each piece of magnetocaloric material in the AMR is cooled down.
- Fluid flow** from the hot external heat exchanger (HHEX) through the cooled magnetocaloric material (where the fluid is cooled down) into the cold external heat exchanger (CHEX) where the heat is absorbed from the surroundings (the refrigerator's chamber).

As presented in Yu et al. [5], 41 prototypes of magnetic refrigerator working at room temperature have been built up to 2010. The existing prototypes show a maximum cooling capacity of up to 1 kW (for a zero temperature span) and a maximum temperature span up to 30 K (for a zero cooling capacity). Although it may be sometimes hard to distinguish, magnetocaloric devices can generally be divided into two groups: prototypes for potential market applications and experimental devices primary used for the testing of various types of active magnetic regenerators (AMRs). Compared to other magnetic-refrigerator prototypes, experimental devices usually operate with a linear motion and are more robust. They must enable the fast and simple variation of different operating

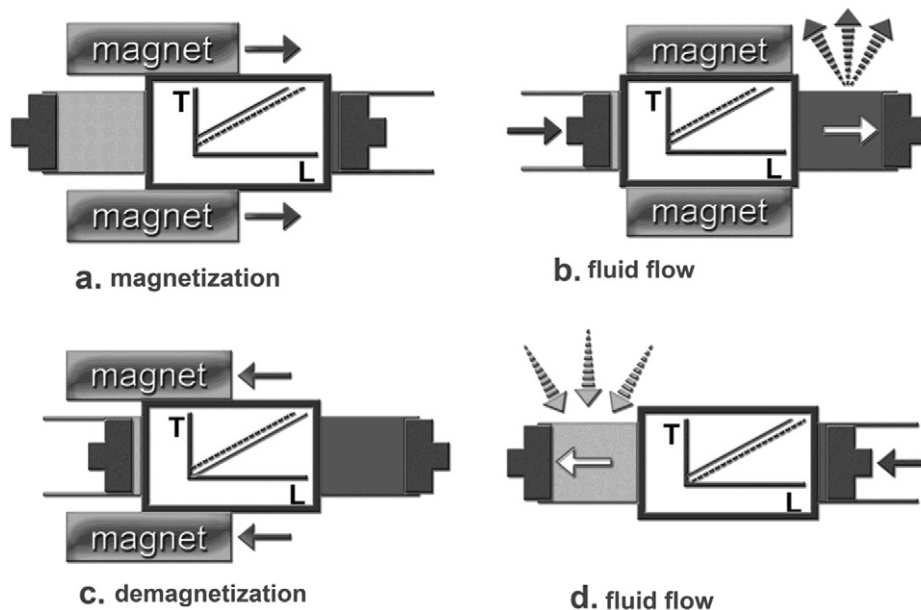


Fig. 1. Schematic of the AMR process.

conditions and different geometries of AMRs. They contain a relatively small amount of magnetocaloric material, usually in the form of one or in some cases two AMRs (due to the magnetic force balance and the continuous operation). On the other hand, prototypes for potential market applications are generally rotary (e.g., the rotation of several AMRs through a static magnetic field), more compact and contain a larger amount of magnetocaloric material. They usually operate at higher frequencies (number of thermodynamic cycles per unit of time) and consequently have higher cooling capacities. Currently, the best cooling characteristics have been shown by the prototypes developed by Engelbrecht et al. [6] and Russek et al. [7].

In recent years several AMR experimental devices have been built and tested [8–15]. These devices apply different permanent-magnet assemblies and the linear movement of the magnet and/or the AMR. Their operation is based on a relatively low operating frequency (1 Hz or below) and a low cooling capacity. The exception among the above-mentioned devices is the one developed by Tura and Rowe [14]. This device does not apply a linear motion but uses the rotation of a nested Halbach assembly, which allows a higher operating frequency (up to 4 Hz). A low frequency, which is mostly due to the long (de)magnetization time, is one of the major disadvantages of all the linear AMR experimental devices. The aim of applying a linear device is therefore related to the characterization of the optimum operating parameters (mass-flow rate and fluid-flow period duration) and measurements of the maximum temperature span, which are not so much effected by the (de)magnetization time.

1.1. The aim of the study

In this paper, a new prototype of magnetic refrigerator, designed as an experimental device for testing the AMR, is presented. It is based on the reciprocating (linear) movement of a permanent-magnet assembly over the static AMR.

The main goal of this study was an experimental comparison of six different active magnetic regenerators (AMRs) with gadolinium as the magnetocaloric material (shown in Fig. 5). The analyzed AMRs can be divided into two groups, parallel-plate and packed-bed AMRs. For the purpose of the analysis of the parallel-plate AMRs, three different AMRs were designed and built in order to study the impact of the spacing between the plates (porosity) and the orientation of the plates in the magnetic field. An innovative technique based on laser welding technology was used for the construction of the parallel-plate AMRs. In the case of the packed-bed AMR the aim of the study was to analyze the impact of the size and the shape of the gadolinium particles. The analyzed packed-bed AMRs were filled with cylinders, powders and spheres.

Furthermore, the operation of the AMR is strongly affected by the mass-flow rate of the heat-transfer fluid through the AMR and by the operating frequency (defined as the number of thermodynamic cycles of the AMR per unit of time). Therefore, the AMRs were measured for a variety of values for those two variables in order to find the optimum corresponding mass-flow rate and the operating frequency for each analyzed AMR.

The AMRs (under different operating conditions) were compared with regard to the maximum established temperature span, the generated cooling capacity and the experimentally predicted COP. This is the most comprehensive experimental analysis of different AMRs performed in a magnetic refrigerator so far.

2. The design of the experimental device and its operation

The experimental device is designed for quick and simple changes to a variety of operating conditions, such as the mass-flow

rate and the operating frequency. It also enables quick and simple changing of the AMR in the device, in order to analyze the impact of the magnetocaloric material and its geometry in the AMR.

The scheme and a photograph of the experimental device are shown in Fig. 2. The device consists of three basic components: the core of the device with the AMR, the magnet assembly and the fluid-manipulation system.

The AMR is placed in the middle of the core part of the device, as shown in Fig. 3. On one side, next to the AMR, the heat exchanger is located. It is constructed as a straight-tube heat exchanger. It transfers heat between the working fluid of the AMR and the external fluid, which is preheated (or cooled) to the desired temperature in the thermostatic chamber. Its function is to provide a constant temperature of the working fluid on one side of the AMR (during operation it is referred to as the hot side). On the other side of the AMR the electric heater (resistive heating wire) is located. It works as a cooling-load simulator and is controlled by a variable transformer. The housing of the core part is made of plastic (Ertalite).

The magnet assembly, which is explained in detail in the next subsection, is based on four Nd–Fe–B permanent magnets and a soft ferromagnetic material that guides the magnetic flux into the air gap of the assembly. The processes of magnetization and demagnetization of the AMR are provided by the movement of the magnet assembly, which is moved and controlled by a pneumatic cylinder.

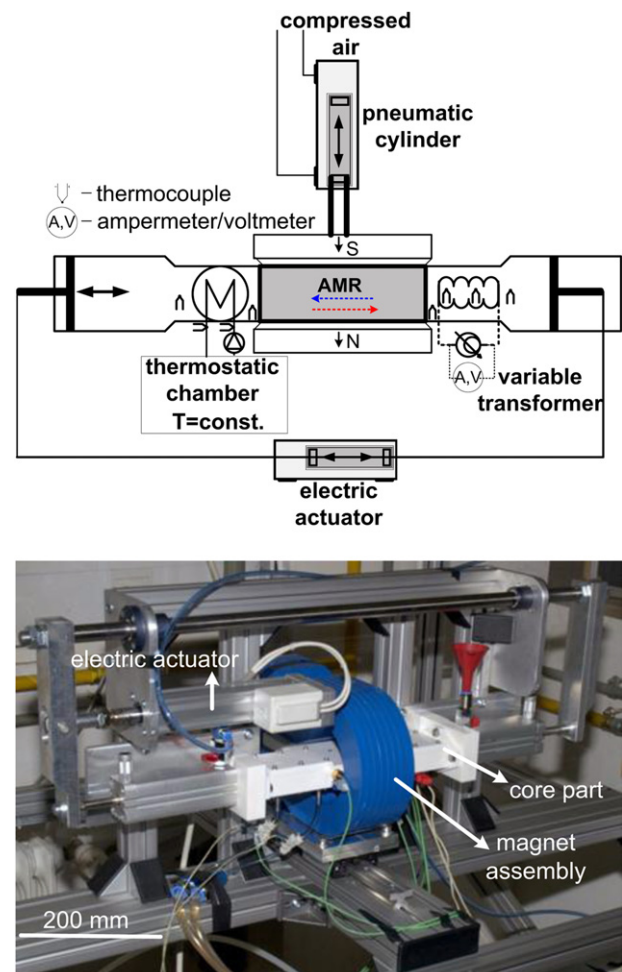


Fig. 2. The scheme (top) and a photograph (bottom) of the experimental device.

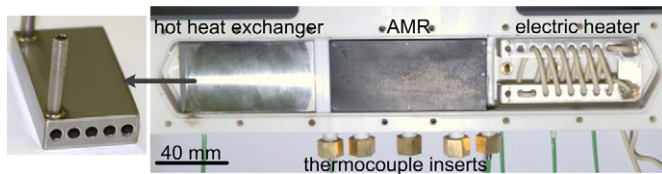


Fig. 3. Photograph of the core part of the experimental device.

The counter-flow operation of the AMR is assured by the fluid-manipulation system. It is based on an electric actuator with an adjustable speed and stroke length. The actuator is further connected to the two pistons in the remanufactured pneumatic cylinders, which are located at both ends of the core part of the device. The movement of the electric actuator therefore moves the pistons, which causes the fluid to flow through the core part and through the AMR.

The experimental device is controlled by LabView software. In the case of the Brayton AMR cycle, the experimental device performs the following four steps, which are in general the same as shown in Fig. 1:

- Magnetization (movement of the magnet assembly over the AMR);
- Fluid flow through the AMR from the cold to the hot side (movement of the electric actuator);
- Demagnetization (movement of the magnet assembly from the AMR);
- Fluid flow through the AMR from the hot to the cold side (movement of the electric actuator).

The temperature span of the AMR is measured with K-type thermocouples, which are placed at both ends of the AMR (between the AMR and the heat exchanger and the electric heater).

2.1. Design of the magnet assembly

The magnet assembly used in the experimental device is based on four Nd₂Fe₁₄B-type permanent magnets with a maximum energy product of 50 MGOe and a soft ferromagnetic material (Steel 37), which guides and focuses the magnetic flux into the air gap. The assembly was developed and optimized using two different types of numerical software.

In the first step, the numerical program FEMM was used to perform 2D numerical simulations and to obtain basic knowledge about the features of the magnet assembly. The geometry and dimensions of the elementary elements of the assembly (the width and the height of the magnet, the thickness of the external steel ring, the height of the air gap and the angle between the magnets) were optimized and adopted according to the obtained results. The 2D numerical simulations were further tested with the more sophisticated ANSYS Multiphysics software, which solves three-dimensional problems. The results gave a basis for the final design and construction of the magnet assembly.

Detailed measurements were performed to experimentally verify the magnetic flux and its distribution in the assembly. This was done by using a three-axis magnetometer with an integrated three-axis Hall probe. In the air gap (magnetization area), the magnetic flux density is 1.15 T, while in the demagnetization area the magnetic flux density is 0.007 T, as shown in Fig. 4 (bottom). Fig. 4 (bottom) also shows a comparison of the numerically calculated and the measured values of the magnetic flux density. The results reveal very good agreement between the measured and the calculated values with the ANSYS Multiphysics numerical program.

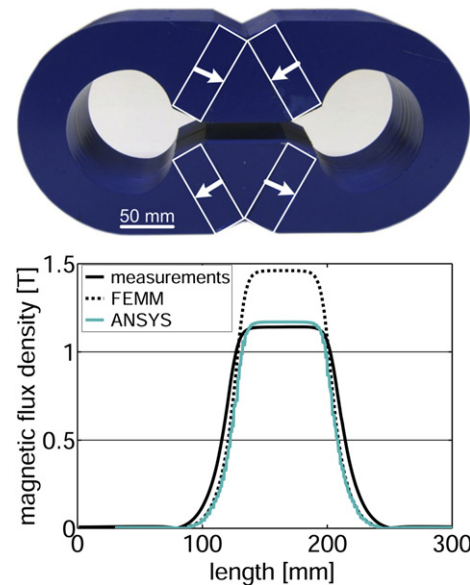


Fig. 4. Photograph of the magnet assembly (top) and a comparison of the measured and numerically calculated values of the magnetic flux density (bottom).

On the other hand, the calculated values from the FEMM program differ considerably from the measured ones. It can be concluded that 2D simulations (FEMM) are not sufficient to perform accurate simulations of the magnetic field. However, such a simple program enables a much faster verification of the basic design concepts.

On the basis of the measured values of the magnetic flux density and the geometry of the magnet assembly, the factor λ_{cool} proposed by Bjoerk et al. [16] was calculated. Its value is 0.064. However, the potential of this assembly is larger. Namely, if in the experimental device two AMRs were to be used, which means that the air gap would always be filled with the magnetocaloric material, the factor λ_{cool} would be twice as large. This would be one of the highest values regarding the comparison of magnet assemblies published by Bjoerk et al. [16].

3. Experimental results and discussion

3.1. Analyzed AMR geometries and operating parameters

As already explained in the introduction, the goal of the experimental study was a comparison of the AMRs, with gadolinium as the magnetocaloric material. The analysis was based on four basic geometries of gadolinium: parallel plates, spheres, powders (in the form of flake) and cylinders. Generally, the AMRs can be divided into two groups: ordered structures (parallel plate) and packed-bed structures (spheres, powder and cylinders). Among the parallel-plate AMRs, three different geometries were tested. Two of them have the plates oriented parallel to the magnetic field, while one was formed with the plate's orientation perpendicular to the magnetic field. The distances between the gadolinium plates in the parallel plate AMRs were set by narrower plates (acting as distance spacer), which were laser welded [17] along the both longer sides of each gadolinium plate. These were further welded into the finite AMR structure, shown in Fig. 5 (A., B., C.). This technique is explained in detail in Tušek et al. [18].

All the analyzed AMRs are shown in Fig. 5, while their geometric properties are presented in Table 1.

All the AMRs shown in Fig. 5 were inserted into the experimental device and measured for their maximum temperature span

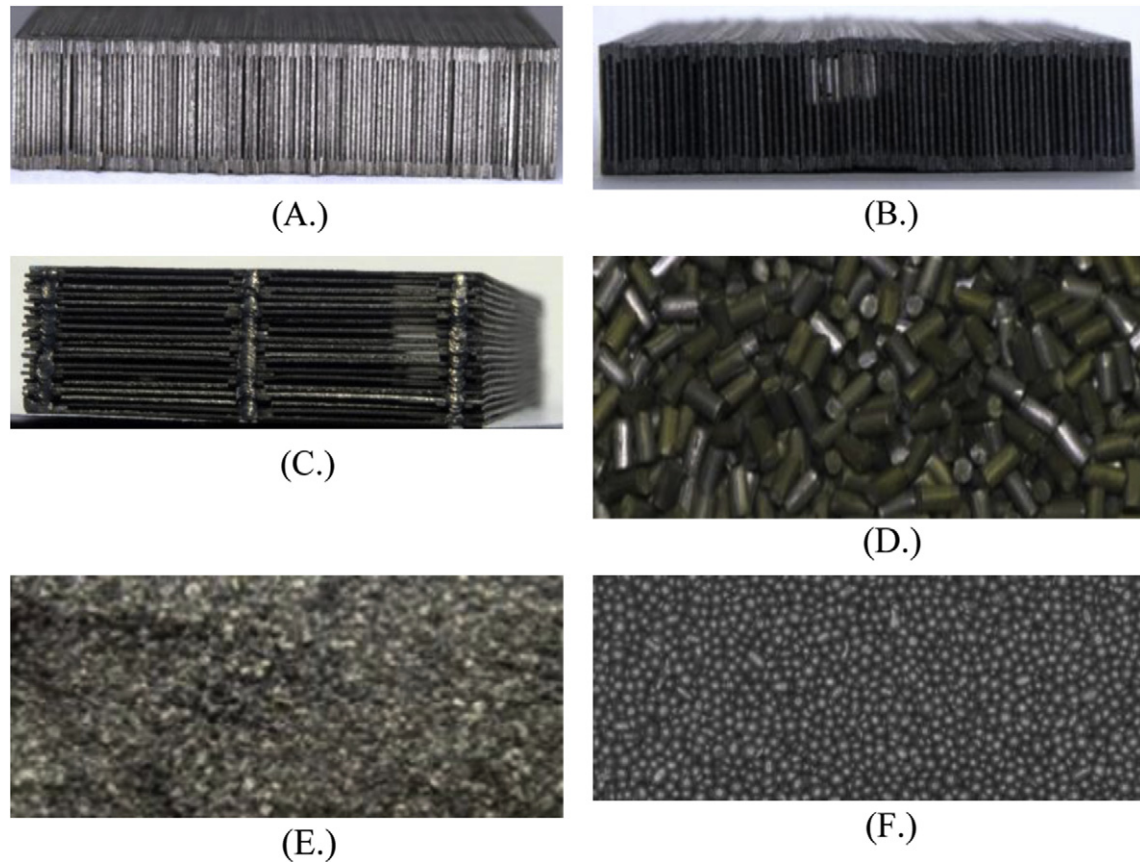


Fig. 5. Photographs of analyzed AMRs (A. – parallel plates with spacing of 0.1 mm; B. – parallel plates with spacing of 0.25 mm; C. – perpendicular plates; D. – cylinders; E. – powder; F. – spheres).

(at a 0 W cooling capacity) at different mass-flow rates and at a constant operating frequency. Moreover, the three AMRs that showed the best characteristics (A., B. and F.) were further tested at different operating frequencies and at different applied cooling capacities.

The experimental parameters are presented in Table 2. It should be noted that the time needed for the (de)magnetization process is 0.75 s (time for the magnet's movement) regardless of the operating frequency. This means that the operating frequency was influenced or changed only by the duration of the fluid-flow period. In addition, the operating frequency is also affected by the time needed to control the electric actuator (piston) and the pneumatic cylinder (magnet assembly) or the so-called dead time. The dead time during one cycle is 0.9 s.

The temperature in the hot heat exchanger was set to 300 K, while the ambient temperature was maintained at 295 K for all the performed tests. However, due to the non-ideal heat transfer in the hot heat exchanger, the temperature on the hot side of the AMR exceeds the hot heat exchanger's temperature under certain operating conditions (e.g., low mass-flow rates).

As the heat transfer fluid a mixture of distilled water and commercial coolant based on ethylene-glycol was used in order to minimize the corrosion of gadolinium.

The key measurement parameters and the variables of the experiment are the temperature span of the AMR (ΔT), the applied cooling capacity ($\dot{Q}_c$) and the mass-flow rate of the fluid flow ($\dot{m}_f$). It is therefore important to estimate its measurement uncertainties. Since all of those three parameters are determined indirectly from

Table 1
Geometry of analyzed AMRs.

Outer dimensions [mm]	(A.)	(B.)	(C.)	(D.)	(E.)	(F.)
	10 (height) $\times$ 80 (length) $\times$ 39 (width)					
Material geometry	Plates; $t = 0.25$ mm	Plates; $t = 0.25$ mm	Plates; $t = 0.25$ mm	Cylinders; $d = 2.5$ mm; $L = 4$ mm	Powder; $d = 0.35\text{--}0.5$ mm	Spheres; $d = 0.35\text{--}0.5$ mm
Plate's distance [mm]	0.1	0.25	0.25	/	/	/
Total heat transfer area [m ²]	0.1395	0.0896	0.0806	0.0198	0.1650	0.2400
Total mass of Gd [kg]	0.1763	0.1309	0.1427	0.1205	0.0930	0.1350
Average porosity [/]	0.2564	0.4667	0.4247	0.4188	0.5515	0.3490
Hydraulic diameter($d_h = 4 \cdot V_{AMR} \cdot \epsilon / A_{ht}$) [mm]	0.2	0.5	0.5	2.22	0.3510	0.1530

Table 2
Experimental parameters.

Operation frequency [Hz]	0.155–0.333
(De)Magnetization time [s]	0.75
Fluid-flow period [s]	0.25–2
Dead time per cycle [s]	0.9
Mass-flow rate [kg/s]	0.0008–0.04
Cooling capacity [W]	0–7
Ambient temperature [°C]	22
Hot heat-exchanger temperature [°C]	27
Heat-transfer fluid	67% of distilled water + 33% commercial coolant based on ethylene-glycol

the measurements, its relative combined measuring uncertainty was estimated. The temperature span is defined as the average temperature difference between the hot and the cold side of the AMR in the steady-state condition, the applied cooling load is defined as the product of the applied current and voltage, which was measured with a multi-meter, while the mass-flow rate was determined based on the movement of the piston and its dimensions. It should be pointed out that the used thermocouples were additionally calibrated in the temperature range between 270 and 300 K. Since the number of repeated experiments was too small for an evaluation of the statistical uncertainty (uncertainty of type A) we could only evaluate the uncertainty of type B, which is based on the specifications of the accuracy of the measuring equipment. A uniform (rectangular) probability distribution was assumed. The relative combined measurement uncertainty of the temperature span, the cooling capacity and the mass-flow rate were estimated to be 1.5%, 1.1% and 2.4%, respectively. These values were calculated based on the selected (assumed) measured values of the temperature span ($\Delta T = 10$ K), the cooling capacity ($\dot{Q}_c = 5$ W) and the mass-flow rate ($\dot{m}_f = 0.1$ kg/s).

3.2. Maximum temperature span

Fig. 6 shows the envelope of the temperature span between the fluid at the hot and the cold sides of the AMR (A.) for the 0 W cooling capacity. Even though the temperature span of the AMR for the 0 W cooling capacity is not relevant for real cooling applications, it is a precondition for the application of the AMR in the magnetic refrigerator.

The same tests as presented in Fig. 6 were subsequently performed under different operating conditions and for different AMRs. Fig. 7 shows the maximum temperature span as a function of the ratio of displaced fluid volume (V^*) for four different operating frequencies and three AMRs (A., B. and F.). The ratio of the displaced fluid volume is defined as the ratio of the fluid volume pumped

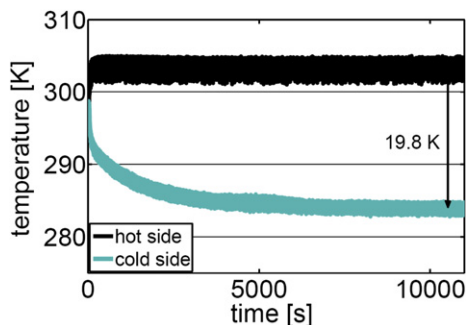


Fig. 6. The envelope of the temperature span at 0 W of cooling capacity ($V^* = 0.365$ and $\nu = 0.3$ Hz).

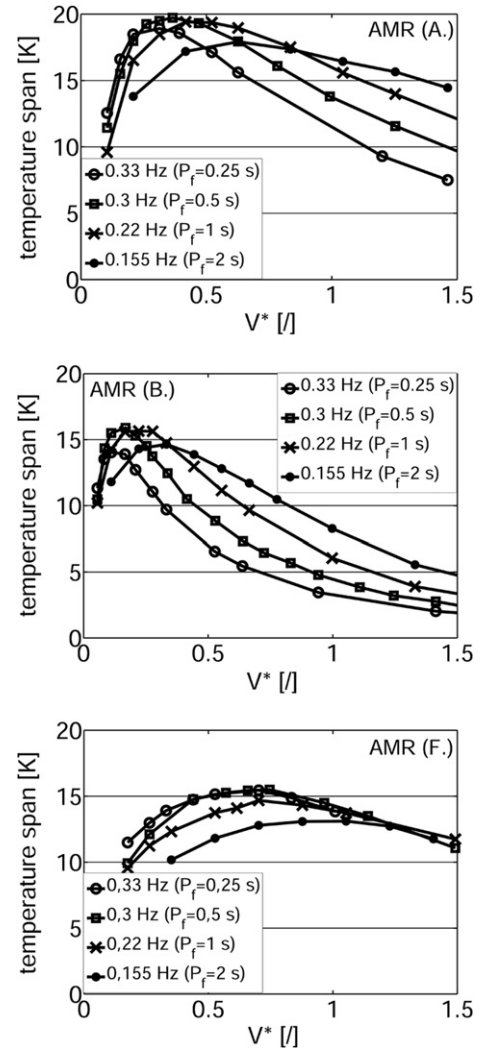


Fig. 7. Temperature span as a function of the ratio of displaced fluid volume at different operating frequencies for the AMR (A., B. and F.).

through the AMR in one fluid-flow period (P_f) with the volume of the AMR occupied by the fluid:

$$V^* = \frac{\dot{m}_f \cdot P_f}{\rho_f \cdot V_{AMR} \cdot \epsilon} \quad (1)$$

It can be concluded that there is an optimum ratio of the displaced fluid volume and consequently an optimum mass-flow rate, which leads to the largest temperature span at each operating frequency. Similar results were previously shown using some numerical simulations of the AMR's operation by e.g. Petersen [19] as well as by some experimental analyses by e.g. Engelbrecht et al. [20]. The reasons for such trends are as follows. If the value of V^* is too small, only a part of the energy generated by the magnetocaloric effect can be utilized. If, on the other hand, the value of V^* is too large, the fluid will utilize the entire energy available from the magnetocaloric effect (depending also on the heat-transfer properties between the fluid and the solid). Further fluid flow (in the same fluid-flow period) causes the fluid from the hot side of the AMR or even from the hot reservoir to flow to the cold reservoir. This leads to a decrease in the temperature span. Furthermore, the optimum values of V^* are strongly related to the AMR geometry and

its heat-transfer area. The larger the heat-transfer area is, the larger the optimum values for V^* will be for each operating frequency.

It is further evident from Fig. 7 that there is an optimum operating frequency (at which the AMR can generate the largest temperature span) for each AMR. For parallel-plate AMRs (A. and B.) the optimum frequency is between 0.22 and 0.3 Hz (only for this experimental device), which means that the largest temperature span is obtained at between 0.5 and 1 s of the fluid period. The spacing between the plates (in the case of equal plate thicknesses) apparently does not affect the optimum frequency, since the thickness of the magnetocaloric material mostly affects the time needed for the fluid to absorb the energy generated with the magnetocaloric effect. For the packed-bed AMR (F.) the optimum frequency is slightly larger than for the parallel-plate AMRs. The main reason for this is the better heat-transfer characteristics, which means that a shorter time is needed for the fluid to absorb the energy generated by the magnetocaloric effect.

Fig. 8 shows a comparison of the maximum temperature span (at 0 W of cooling capacity) as a function of V^* for all six analyzed AMRs. The results in Fig. 8 were obtained for an operating frequency of 0.3 Hz ($P_f = 0.5$ s). The largest temperature span (19.8 K) was obtained with the AMR (A.) that is made of parallel plates with a spacing of 0.1 mm. The AMR (B.) was made of parallel plates with a spacing of 0.25 mm. This led to a smaller temperature span compared to the AMR (A.), mostly due to the smaller heat-transfer area and the larger hydraulic diameter. The smallest temperature span among the parallel-plate AMRs was measured with the AMR (C.). In this case, the parallel plates have a spacing of 0.25 mm, but unlike AMR (B.), the plates are oriented perpendicular to the magnetic field. The main reason for such a difference can be attributed to the demagnetization effect (caused by the demagnetization field as a consequence of the magnetization of the magnetic material). The demagnetization field is opposing the magnetization in the material; it reduces the internal magnetic field and consequently the magnetocaloric effect. Recently, Nielsen et al. [21] have analyzed the effect of the magnetocaloric material's geometry on the magnetocaloric effect. They concluded that the geometry of the material significantly affects the internal magnetic field in the material. In the case of the AMR with plate's orientation perpendicular to the magnetic field the generated cooling capacity can be up to 40% smaller than with the plate parallel to the magnetic field.

Among the packed-bed AMRs the largest temperature span was generated with the AMR (F.) filled with spheres. A considerably lower temperature span was achieved with the AMR filled with powder (E.) and cylinders (D.). As it is also clear from Fig. 8, the packed-bed AMRs generated smaller maximum temperature spans compared to the parallel-plate AMRs, even though they have a larger heat-transfer area and a smaller hydraulic diameter (Table 1). One of

the reasons for this can be attributed to the non-uniform porosity of the packed-bed regenerators. It is important to point out that near the regenerator's wall, the porosity is much larger than in the middle of the regenerator. This results in a non-uniform fluid flow [22] and, consequently, a lower regenerator efficiency. This effect is more pronounced in the case of a relatively narrow regenerator (which to a large extent applies to our AMR dimensions) filled with relatively large particles [23]. The non-uniform porosity is, in the case of the spheres and powder, additionally affected by the uneven particle distribution (between 0.35 and 0.5 mm). The shape of the particles in the regenerator significantly affects the uniformity of the porosity as well. The powder and cylinders have, in addition to the spheres, more than one spatial degree of freedom. Therefore, they can be randomly oriented inside the regenerator. This further means that the local porosity in the regenerator is not homogenous and can differ drastically from the average porosity. This is the main reason why the AMR filled with powder (E.) achieves a much smaller temperature span compared to the AMR with the spheres (F.), even though their dimensions are relatively similar. In the case of the AMR (D.) that is filled with the cylinders the additional problem is the relatively large size of the cylinders. This further increases the problem of non-uniform porosity and decreases the heat-transfer performance (a smaller heat-transfer area and a larger hydraulic diameter). Consequently, the AMR (D.) generated the smallest temperature span, as shown in Fig. 8.

3.3. Temperature span at different cooling capacities

The three AMRs (A., B. and F.) that achieved the largest maximum temperature span were further tested by adding a cooling capacity at the cold reservoir (through the electric heater). Fig. 9 shows the envelope of the temperature span between the fluid at the hot and the cold sides of the AMR (A.) for different applied cooling capacities. As expected, the temperature span decreases with the increased cooling capacity.

Fig. 10 shows the comparison of the three AMRs at three different values of V^* and at a constant operating frequency (0.3 Hz). An almost linear dependence of the cooling capacity and the temperature span (cooling line) was observed. It is evident that the larger is the ratio of the displaced fluid volume, the greater the inclination of the slope of the cooling line will be, since the cooling capacity is in general increasing with the mass-flow rate. A large ratio of the displaced fluid volume (V^*) therefore leads to a high cooling capacity, but it is limited to a smaller temperature span. It is further evident that AMR (B.) shows the best characteristics at lower values of V^* , since it can generate the largest temperature span under such operating conditions (see Fig. 7). With the increase of V^* the performance of the AMR (B.) is decreasing (due to the limited temperature span at larger values of V^*). At higher values of V^* the best characteristics can be achieved with the AMR (A.). This is

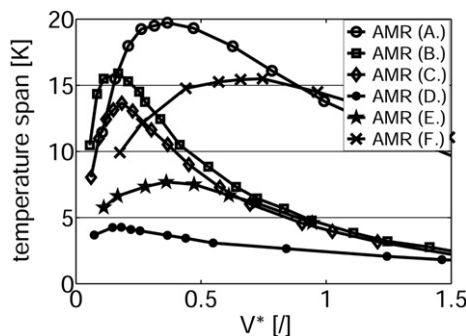


Fig. 8. Temperature span as a function of the ratio of displaced fluid volume for all six analyzed AMRs in the case of an operating frequency of 0.3 Hz ($P_f = 0.5$ s).

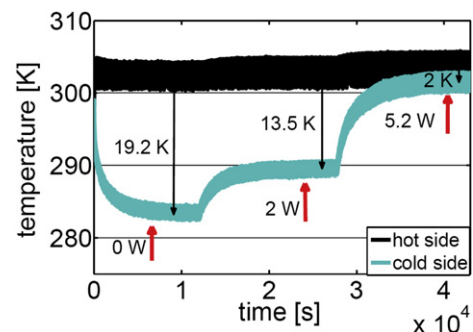


Fig. 9. The envelope of the temperature span at different cooling capacities ($V^* = 0.418$ and $\nu = 0.3$ Hz).

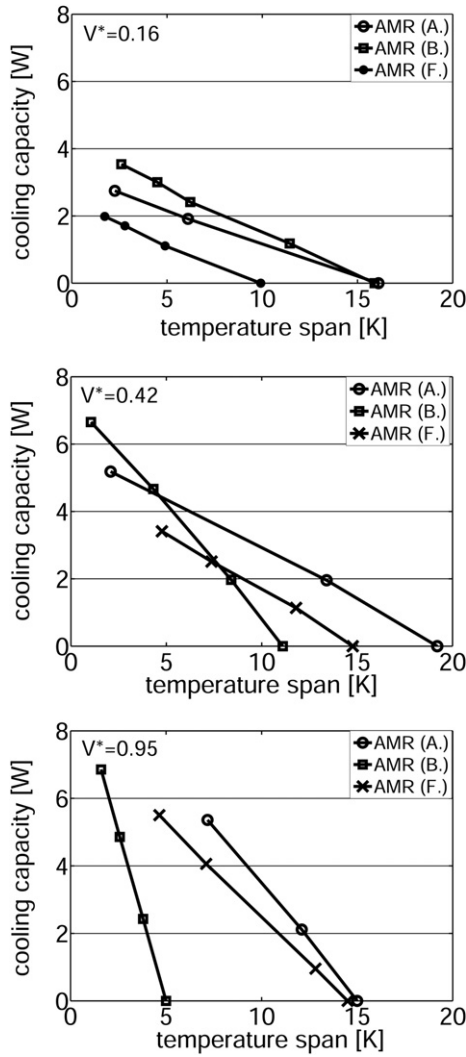


Fig. 10. Cooling capacity as a function of temperature span (cooling line) at different values of the ratio of displaced fluid volume for three AMRs (A., B. and F.).

mostly due to the larger heat-transfer area, which means that a relatively large temperature span is generated at higher values of V^* . This is further related to a larger cooling capacity. Moreover, it is evident that the AMR (B.) has the largest slope of the cooling line. This is due to the higher porosity of this AMR compared to the AMR (A.) and the AMR (F.). This means that at the same value of V^* the mass-flow rate through the AMR (B.) is larger than through the other two AMRs. This consequently increases the cooling capacity.

It should be noted that the measured cooling capacities are relatively small. This is mostly due to long (de)magnetization times and, consequently, the small operating frequencies. If the analyzed AMRs were to be used in, e.g., a rotating device with a (de)magnetization time below 0.1 s and zero dead time, the measured cooling capacities would be up to five times larger (due to the five times higher frequency).

The heat losses to the surroundings represent an additional loss mechanism that decreases the AMR's performance in the experimental device. The thermal resistance of the device can be divided into two parts. The part of the device where the AMR is inserted is not isolated, since the height of the air gap in the magnet assembly is fitted to the outer dimensions of the housing of the device, which prevents any isolation. The thermal resistance of this part of the device is estimated to be 9 K/W. All the other parts of the device were isolated and its thermal resistance was estimated to be 15 K/W.

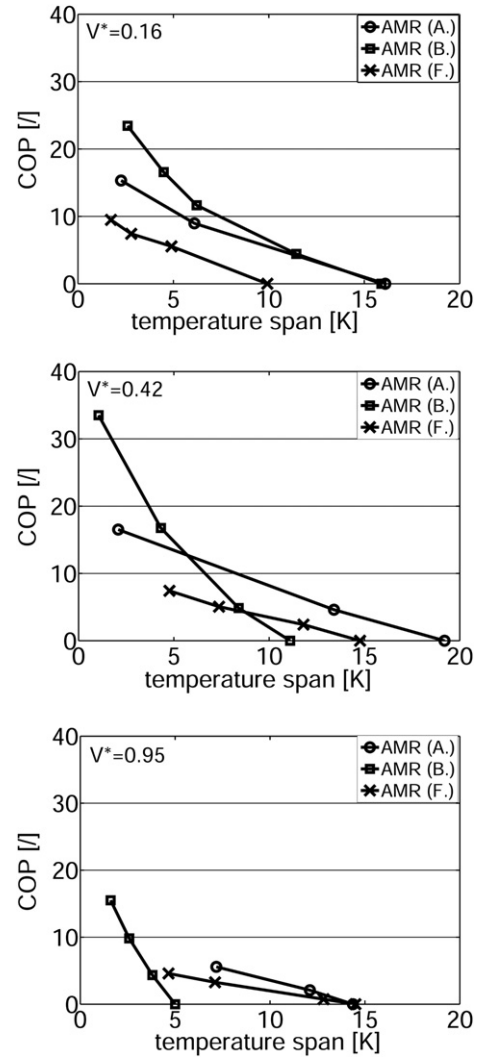


Fig. 11. Experimentally predicted values of the COP as a function of the temperature span for different values of the ratio of the displaced fluid volume for three AMRs (A., B. and F.).

3.4. Experimentally predicted COP

Based on the measured results presented in Fig. 10, the COP values of the three AMRs (A., B. and F.) were analyzed. Since the input magnetic work was not measured during the experiment, it was calculated instead. This was done using the numerical model presented by Tušek et al. [24]. The model was validated with the experimental results as presented in Tušek et al. [25]. In order to generalize the results the model neglects the mechanical losses (motor inefficiency) of the driving system. The work required for the fluid pumping through the AMR was calculated using the friction-factor correlations. The measured values of the cooling capacity and the temperature span were used as the input parameters for the numerical model.

The COP is defined as the ratio between the cooling load and the sum of the input work, which can be further divided into the work required for the pumping of the fluid and the magnetic work. The latter, expressed in terms of power, is defined as:

$$\dot{W}_{\text{mag}} = m_s \cdot v \cdot \oint T \cdot ds = m_s \cdot v \cdot \int_0^L \int_0^{2P} T(x, t) \cdot \frac{ds(x, t)}{dt} dt dx \quad (2)$$

The power of the magnetic work is defined as the integral of the temperature with respect to the entropy for each segment of the magnetocaloric material in the AMR. The integration is performed over the entire cycle in the steady-state condition. Next, the integrated value is multiplied by the total mass of the magnetocaloric material (m_s) and the operating frequency (ν). Such a procedure was proposed by Engelbrecht [26] and Šarlah [27].

The power of the work required to pump the fluid depends on the mass-flow rate of the fluid ($\dot{m}_f$), the fluid density (ρ_f), the pump efficiency, which was estimated to be 0.8, and the pressure drop. The latter was calculated using the well-known equation presented in, e.g., Kays and London [28]. It should be noted that the calculated pressure drop includes only the friction in the core of the regenerator, since the entrance, exit and flow acceleration effects can be neglected for regenerators with small hydraulic diameters [28]. The friction factors for the packed-bed and parallel-plate structures are defined using the correlations published in Kays and London [28].

The results of the *COP* analysis are presented in Fig. 11 as a function of the temperature span for three AMRs (A., B. and F.). These are shown at three different values of V^* and at a constant operating frequency (0.3 Hz; $P_f = 0.5$ s). Generally, the trends of the *COP* are relatively similar to the trends of the cooling capacity (Fig. 10). At larger temperature spans the most efficient is AMR (A.), while at smaller temperature spans the AMR (B.) has the largest values of the *COP*. This is mostly due to the higher porosity of the AMR (B.), as described in Section 3.3. It should be noted that the higher pressure drop of the AMR (A.) does not significantly affect its efficiency compared to the AMR (B.), which has a larger hydraulic diameter and, consequently, a smaller pressure drop. However, the packed-bed AMR (F.) has the largest pressure drop and thus smaller values of the *COP*. It should be further noted that with the increased V^* the magnetic work is increased as well. This is due to larger overlapping of the magnetic thermodynamic cycles of each piece of magnetocaloric material in the AMR (and the work required to pump the fluid is increased as well). From the *COP* point of view the optimal value of V^* is around 0.42 (with regard to the analyzed operating parameters).

4. Conclusions

A new experimental testing device for AMRs has been designed and built. A comprehensive experimental analysis of six different AMRs was performed in order to analyze the impact of the geometry of the magnetocaloric material in the AMR and the impact of the operational conditions (mass-flow rate and operating frequency). Based on the obtained results the following conclusions can be drawn.

Each AMR has a corresponding optimum operating mass-flow rate and operating frequency (from the maximum-temperature-span point of view). The larger the heat-transfer area is, the higher the optimum mass-flow rate will be. However, a higher operating mass-flow rate increases the cooling capacity as a consequence. It is therefore very important that the optimum mass-flow rate from the temperature-span point of view is as high as possible in order to generate a large cooling capacity for high temperature spans. Furthermore, a higher operating frequency also increases the cooling capacity of the AMR. The optimum operating frequency or optimum duration of the fluid flow through the AMR is closely related to the heat-transfer coefficient in the AMR and/or the thickness of the magnetocaloric plate or sphere (particle). The faster the fluid absorbs all the heat generated by the magnetocaloric effect, the larger the optimum frequency will be. The larger heat-transfer coefficient of the packed-bed AMR contributes to the higher optimum frequency compared to the parallel-plate AMR. Therefore, the AMR should contain as thin a piece of magnetocaloric material as possible, a small hydraulic diameter and as large a heat-transfer area as

possible. These characteristics will also ensure a large cooling capacity for high temperature spans between the heat source and the heat sink of the magnetic refrigerator.

Among the parallel-plate AMRs the best cooling characteristics (temperature span, cooling load and the *COP*) were in general obtained by AMR (A.) (see Fig. 5). This indicates that the parallel-plate AMR should be constructed with a small spacing (0.1 mm or below) and with the plates' orientation being parallel to the magnetic field. AMR (B.) has a larger spacing between the plates and thus poorer heat-transfer properties (smaller heat-transfer area and a larger hydraulic diameter), while AMR (C.) has plates oriented perpendicular to the magnetic field, which results in a larger demagnetization factor and consequently a smaller magnetocaloric effect. However, it is expected that a particular spacing between the plates, below 0.1 mm, would be optimum, since decreasing the spacing causes an increased pressure drop.

Of the packed-bed AMRs the best cooling characteristics were obtained with AMR (F.), which is filled with the spheres. A significantly lower temperature span was established within AMR (D.) and (E.), which were filled with cylinders and powder. The main reason for this is the fact that the powder and the cylinders have more than one spatial degree of freedom. This means that such particles can be randomly oriented in the AMR, which decreases the homogeneity of the particle distribution, as well as the fluid flow voids through the AMR. Consequently, the efficiency of the heat regeneration and the cooling characteristics is decreased substantially.

Generally, the parallel-plate AMRs show better characteristics than the packed-bed structures. This is somewhat surprising since, for example, AMR (F.), filled with spheres, has better heat-transfer properties (a larger heat-transfer area and a smaller hydraulic diameter) than AMR (A.), as presented in Table 2. One of the reasons for that could be related to the larger, nonhomogeneous, particle distribution or fluid voids in the case of spheres compared to the case of plates. Furthermore, an important impact on the operation of the packed-bed AMRs might also be the aluminium holder, which was used as an additional housing for the packed-bed AMRs. It is assumed that the impact of the aluminium wall of the holder decreases the heat-regeneration efficiency along the AMR and consequently its cooling characteristics.

An important conclusion of this work also relates to the new and innovative technology for constructing the parallel-plate AMR. Using laser welding technology, for the first time, a parallel-plate AMR (AMR (A.)) has been constructed. Furthermore, it provided the largest temperature span compared to existing published results on measured temperature spans with parallel-plate AMRs (for a given magnetic field change generated with permanent magnets as a magnetic field source). With this technique it is possible to construct an AMR with any thickness of spacing and a relatively uniform distribution of the plate. This is crucial for the efficient operation of the AMR (see Nielsen et al. [29] for details on flow maldistribution in a parallel-plate AMR). The quality of the constructed parallel-plate AMR is also one of the reasons why the parallel-plate AMRs performed better than the packed-bed AMRs.

If one compares the *COP* values for different AMRs, it can be concluded that the parallel-plate AMRs can operate also with higher efficiency, since the influence of the pressure drop in the packed-bed AMRs is significantly lower. For the construction of an energy-efficient magnetic refrigerator, the focus should be on the parallel-plate AMRs or other types of ordered structures.

References

- [1] E. Warburg, Magnetic investigations (in German), Ann. Phys. Chem. 13 (1881) 141–164.
- [2] G. Brown, Magnetic heat pumping near room temperature, J. Alloy Compd. 47 (1976) 3673–3680.

- [3] V.K. Pecharsky, K.A. Gschneidner Jr., Giant magnetocaloric effect in $\text{Gd}_5(\text{Si}_2\text{Ge}_2)$, *Phys. Rev. Lett.* 78 (1997) 4494–4497.
- [4] E. Bruck, O. Tegus, D.T.C. Thanh, T. Nguyen, T. Trung, K.H.J. Bushow, A review on Mn based materials for magnetic refrigeration: structure and properties, *Int. J. Refrig.* 31 (2008) 763–770.
- [5] B. Yu, M. Liu, P.W. Egolf, A. Kitanovski, A review of magnetic refrigerator and heat pump prototypes built before the year 2010, *Int. J. Refrig.* 33 (2010) 1029–1060.
- [6] K. Engelbrecht, D. Eriksen, C.R.H. Bahl, R. Bjørk, J. Geyti, J.A. Lozano, K.K. Nielsen, F. Saxild, A. Smith, N. Pryds, Experimental results for a novel rotary active magnetic regenerator, *Int. J. Refrig.* 35 (2012) 1498–1505.
- [7] S. Russek, J. Auringer, A. Boeder, J. Chell, S. Jacobs, C.B. Zimm, The performance of a rotary magnet magnetic refrigeration with layered beds, in: *Thermag IV Conference Proceedings*, 2010, pp. 339–349.
- [8] T. Kawanami, S. Hirano, K. Fumoto, S. Hirasawa, Evaluation of fundamental performance on magnetocaloric cooling with active magnetic regenerator, *Appl. Therm. Eng.* 31 (2011) 1176–1183.
- [9] C.R.H. Bahl, T.F. Petersen, N. Pryds, A. Smith, A versatile magnetic refrigeration test device, *Rev. Sci. Instrum.* 79 (2008).
- [10] S. Bour, J.L. Hamm, H. Michot, C. Muller, Experimental and numerical analysis of a reciprocating room temperature active magnetic regenerator, in: *Thermag III Conference Proceedings*, 2009, pp. 415–424.
- [11] L.A. Tagliafico, F. Scarpa, F. Valsuani, G. Tagliafico, Preliminary experimental result from a linear reciprocating magnetic refrigerator prototype, *Appl. Therm. Eng.* 52 (2) (2013) 492–497.
- [12] C. Dupuis, A.J. Vialle, U. Legait, A. Kedous-Lebouc, D. Ronchetto, New investigation in magnetic refrigerator device, AMR cycle and refrigerant bed performance evaluation, in: *Thermag III Conference Proceedings*, 2009, pp. 437–442.
- [13] P.V. Trevisoli, J.R. Barbosa Jr., R.T.S. Ferreira, Design and preliminary results of a Gd-based linear reciprocating active magnetic regenerator test apparatus, in: *Thermag IV Conference Proceedings*, 2010, pp. 323–332.
- [14] A. Tura, A. Rowe, Permanent magnet magnetic refrigerator design and experimental characterization, *Int. J. Refrig.* 34 (2011) 628–639.
- [15] M. Balli, O. Sari, C. Mahmed, Ch. Besson, Ph. Bonhote, D. Duc, J. Forchelet, A pre-industrial magnetic cooling system for room temperature application, *Appl. Energy* 98 (2012) 556–561.
- [16] R. Bjoerk, C.R.H. Bahl, A. Smith, N. Pryds, Review and comparison of magnet designs for magnetic refrigeration, *Int. J. Refrig.* 33 (2010) 437–448.
- [17] M. Pleterski, J. Tušek, L. Kosec, T. Muhič, M. Muhič, Laser repair welding of molds with various pulse shapes, *Metalurgija (Sisak)* 49 (2010) 41–44.
- [18] Jaka Tušek, K. Pompe, A. Kitanovski, Janez Tušek, A. Poredoš, The manufacturing process of the active magnetic regenerator (in Slovene). Patent Application, P-201200342 (2012).
- [19] T.F. Petersen, Numerical modelling and analysis of a room temperature magnetic refrigeration system, PhD thesis, DTU Mechanical Engineering, Risoe Institute, Denmark, 2007.
- [20] K. Engelbrecht, C.R.H. Bahl, K.K. Nielsen, Experimental results for a magnetic refrigerator using three different types of magnetocaloric material regenerators, *Int. J. Refrig.* 34 (2011) 1132–1140.
- [21] K.K. Nielsen, A. Smith, C.R.H. Bahl, U.L. Olsen, The influence of demagnetizing effects on the performance of active magnetic regenerators, *J. Appl. Phys.* 112 (2012) 094905.
- [22] C.G. duToit, Radial variation in porosity in annular packed beds, *Nucl. Eng. Des.* 238 (2008) 3073–3079.
- [23] D. Nemec, J. Levec, Flow through packed bed reactors: 1. single-phase flow, *Chem. Eng. Sci.* 60 (2005) 6947–6957.
- [24] J. Tušek, A. Kitanovski, I. Prebil, A. Poredoš, Dynamic operation of an active magnetic regenerator (AMR): numerical optimization of a packed-bed AMR, *Int. J. Refrig.* 34 (2011) 1507–1517.
- [25] J. Tušek, A. Kitanovski, U. Flisar, S. Zupan, I. Prebil, A. Poredoš, Active magnetic regenerator (AMR) test device, in: *Thermag V Conference Proceedings*, 2012, pp. 429–436.
- [26] K. Engelbrecht, A numerical model of an active magnetic regenerator refrigerator with experimental Validation, PhD thesis, University of Wisconsin, Madison, USA, 2008.
- [27] A. Šarlah, Thermohydraulic properties of heat regenerators in magnetic refrigerators, PhD thesis, University of Ljubljana, Slovenia, 2008.
- [28] W.M. Kays, A.L. London, *Compact Heat Exchangers*, McGraw-Hill Book Company, New York, USA, 1984.
- [29] K.K. Nielsen, K. Engelbrecht, D.V. Christensen, J.B. Jensen, A. Smith, C.R.H. Bahl, Degradation of the performance of microchannel heat exchangers due to flow maldistribution, *Appl. Therm. Eng.* 40 (2012) 236–247.